



OSP Construction Standards and Procedures

Quality Assurance Guide

Version 1.1

05/06/2024

Change History

Version	Date	Updated By	Update Reason
1.1	05/06/2024	Brittney Everett, Karen Fleming, Phil Schueler, George Lewis	Final Version
1.0	05/02/2024	Josh Cook, Daniel Beckett, Jeff Crabtree, Phil Schueler	Original Version

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Purpose of Guide:

All Kinetic OSP Construction Teams, (albeit Internal or External), will follow the guidelines contained within this document to create standardization and consistency when installing our Kinetic OSP Fiber Optic Network.

This Quality Assurance Guide, (hereinafter, QA Guide), will provide the necessary guidelines for Kinetic OSP Construction crews and External OSP Construction vendors to follow when installing & inspecting ALL OSP Aerial, Underground & Splicing Construction within the Kinetic footprint.

Responsibility

The responsibility to ensure Quality Construction occurs, within our OSP Fiber Network, resides with ALL levels of the Kinetic OSP Construction Team. This includes: OSP Construction Leadership, Project Managers, Construction Inspectors, Construction Analysts & Construction Crews. We are to hold each other accountable to ensure our efforts meet the level of excellence we strive for by focusing on safety, attention to detail and consistency. If we address our work with a One and Done mindset, we will achieve our goals, and our customers will receive the reliable network and quality they expect to see from Kinetic.

Inspection Forms:

OSP Aerial Inspection Form:

**See Appendix A for a Downloadable / Printable .XLSX & .PDF copy.

OSP Aerial Inspection Form			
Date of Inspection:			
Task	Description	Results	Inspection Comments
1	Exchange / Project Name:		
2	Technician Name (First and Last name):		
3	Job Numbers (JobTrac/KC/etc.):		
4	Is the pole placed at the proper depth, plumb, tagged, and restored around the base of the pole (Is the site clean around it, free from excess dirt and rocks)	N/A v	
5	Is the anchor installed correctly - Is the Anchor in the correct location, is it bonded, and is there a guy guard (Is it secured, try to slide it up or down)? Location, Depth, Guy, Lead, B&G, Guards	N/A v	
6	Is the strand to support the cable installed correctly? - Is the down guy bonded to the strand? - Was the strand installed to the proper tension to support the cable sag? Sag, B&G, Clearance, Hardware, Overhead Guy	N/A v	
7	Is there Proper Clearance - Does the height of the cable meet NESC Codes for Clearance?	N/A v	
8	Pole Grounds - If called for on the print (PM2A), is the strand bonded to the pole ground?	N/A v	
9	Was the Correct Hardware used - Strand Vises, 5/8" bolts, curved washers, 3 bolt suspension clamps, 5/8" support Straps	N/A v	
10	Is the cable lashed to the strand correctly - Is the cable straight and not looped over the strand, is the fiber double lashed (Check Lashing, Supports, Spacers, and U Guards are in place, 1 span each direction from the poles selected to be inspected.)	N/A v	
11	Straps, Spacers, and Tags - Were straps and spacers used at the poles, is there a Windstrem 800 # tag on the fiber at the suspension clamp?	N/A v	
12	Is the fiber slack properly stored on Snow Shoes - Are the Snow shoes mounted correctly, is the slack lashed to the strand, and if there is a closure is it mounted correctly to the strand?	N/A v	
13	Is the closure properly secured to the strand, and bonded/grounded? Closures must be secured to the strand, slack must be stored on PM92's (snow shoes) slack must be lashed to strand or secured with Deltec 400 lb. straps	N/A v	
14	Transition from aerial to conduit to FDP (if applicable): Please inspect Risers/ U-guards, trench, pedestals, or any conduit (need to be sealed)	N/A v	
15	Fiber Distribution Panel - Is cable entrance neat? Is the FDP securely mounted, labelled correctly, grounded, and neat?	N/A v	



kinetic

Sensitivity: Internal

CI Conducting Inspection:

CM Conducting Inspection:

Training Rep Present:

Safety Rep Present:

Engineer Present:

Operations Rep Present:

OSP Buried Inspection Form:

**See Appendix A for a Downloadable / Printable .XLSX & .PDF copy.

OSP Buried Inspection Form			
Date of Inspection:			
Task	Description	Results	Inspection Comments
1	Exchange / Project Name:		
2	Technician Name (First and Last name):		
3	Job Numbers (JobTrac/KC/etc.):		
4	Cable - Depth, Location, Restoration (Plow rip must be checked, Fiber must be placed at a minimum of 36" depth unless extra depth is called for on prints, trench or plow rip must be closed up, not sunk, and restored with seed & straw where needed.)	N/A v	
5	Entry/Exit Bore Pits - Are the bore pits backfilled, tamped, and surface restored?	N/A v	
6	Warning Signs - Are marker signs installed straight, 2' in ground with peg?	N/A v	
7	Pedestals/Crossbox - Are pedestals/crossboxes in the correct location, stenciled correctly with orange decals, at the correct depth, and level?	N/A v	
8	Pedestals/Crossbox - Are cables neat and organized, tied down properly, and routed correctly (Loops have enough slack to move and access area behind, tied to crossbar)? Do they have properly labelled cable tags (correct pair count, date, and location)?	N/A v	
9	Pedestals/Crossbox - Is the pedestal/crossbox bonded and grounded properly? (Each cable has it's own ground braid and is attached securely on the bond and the ground bar)	N/A v	
10	Pedestals/Crossbox - Is there the proper amount of pea gravel with a moisture barrier? Is there a dated ant bait with a sticker (where required)? Is there duct seal on conduit coming up into the pedestal?	N/A v	
11	Crossbox - Stencil sheet and cable counts in cross box updated: Stencil sheet and cable counts match print specifications.	N/A v	
12	Handhole - HH Lids - Level and secured, bolts tightened (All bolts should be in cover and tight, HH must be level with surface, set on a 8-10 inches of 2" washed stone)	N/A v	
13	Handhole/Manhole - HH - Gravel, ant bait, splice case bonding & grounding, numbering (Ground rod for splice in HH, ground wire attached to splice case, ground wire terminated to BM531. Ant Bait properly installed with sticker dated where required.) - MH - Bonding & grounding, racks, closures (Splice closure secured to racks and grounded.)	N/A v	
14	Handhole/Manhole - HH - Cable (All fiber coiled up and tagged correctly, bends meet manufacturers bend radius specifications) - MH - Cable - Bends, Racking, Entrances, Securing (All fiber coiled up and tagged correctly, bends meet manufacturers bend radius specifications.)	N/A v	
15	Handhole/Manhole - Conduit HH - Is there duct seal on any conduit coming into HH? - MH - Depth, Location, Pull String, Sealed (Conduits must be sealed, vacant ducts must have pull strings in them.)	N/A v	
16	Manhole - Risers - Bends, Strapping, Guards, Seal, location (MH Fiber Risers require 2, 6" U Guards, duct sealed on riser, all fiber coiled up and tagged, bends must meet manufacturers bend radius specifications.)	N/A v	
17	Fiber Distribution Panel - Is cable entrance neat? Is the FDP securely mounted, labelled correctly, grounded, and neat?	N/A v	
CI Conducting Inspection:			
CM Conducting Inspections:			
Training Rep Present:			
Safety Rep Present:			
Engineer Present:			
Operations Rep Present:			



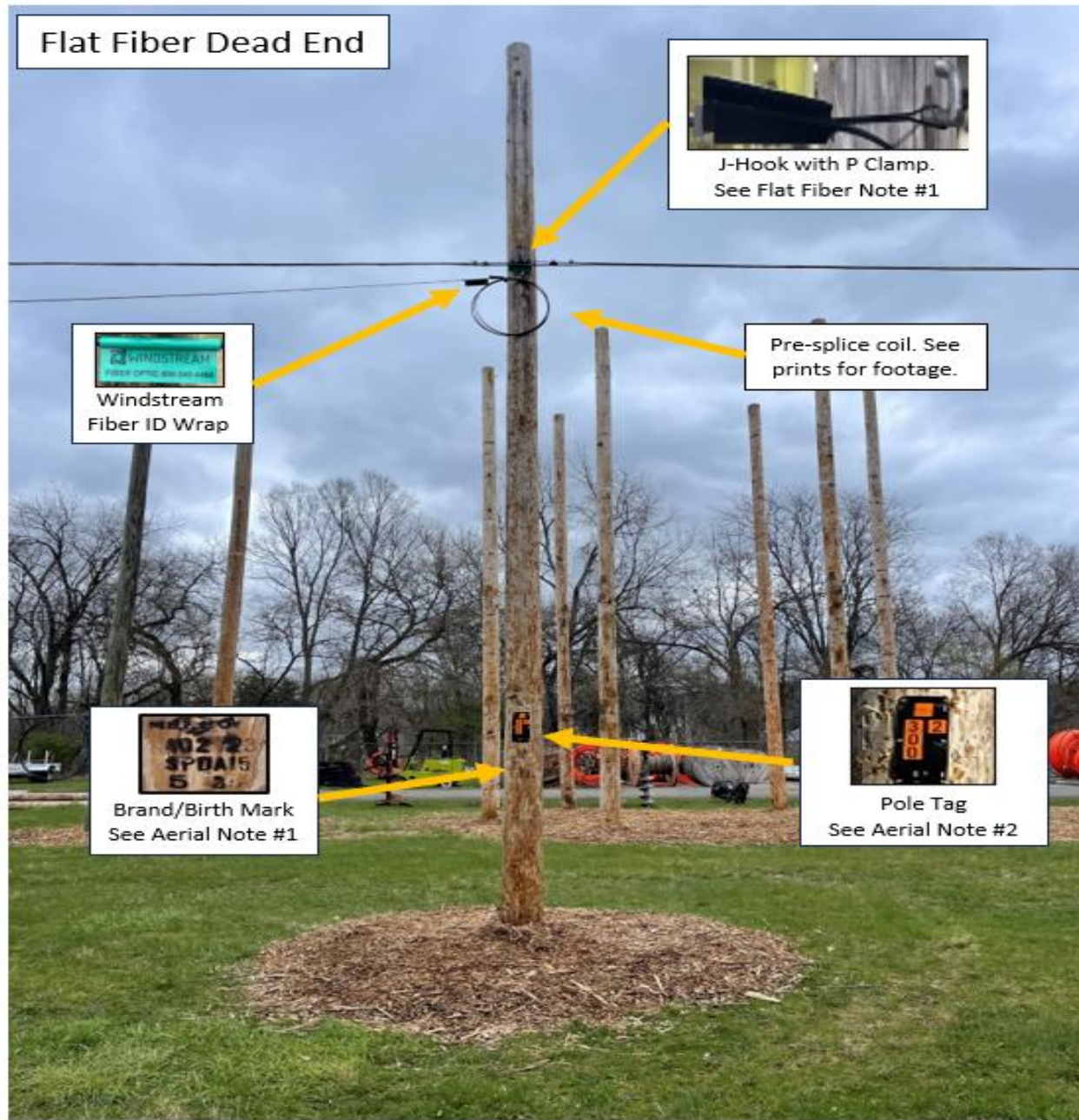
kinetic

Sensitivity: Internal

Aerial Cable Construction:

Aerial Flat Fiber:

Flat Fiber Images



Flat Fiber In-line



J-Hook with (2) P Clamps
See Flat Fiber Note #2



Windstream
Fiber ID Wrap

Pre-splice coil. See
prints for footage.



Brand/Birth Mark
See Aerial Note #1



Pole Tag
See Aerial Note #2

Flat Fiber Notes

1. J-Hook & P-Clamp:

- a. Driven into front of pole and pointed up.
- b. Shim under wedge with fiber cable between it and sleeve.

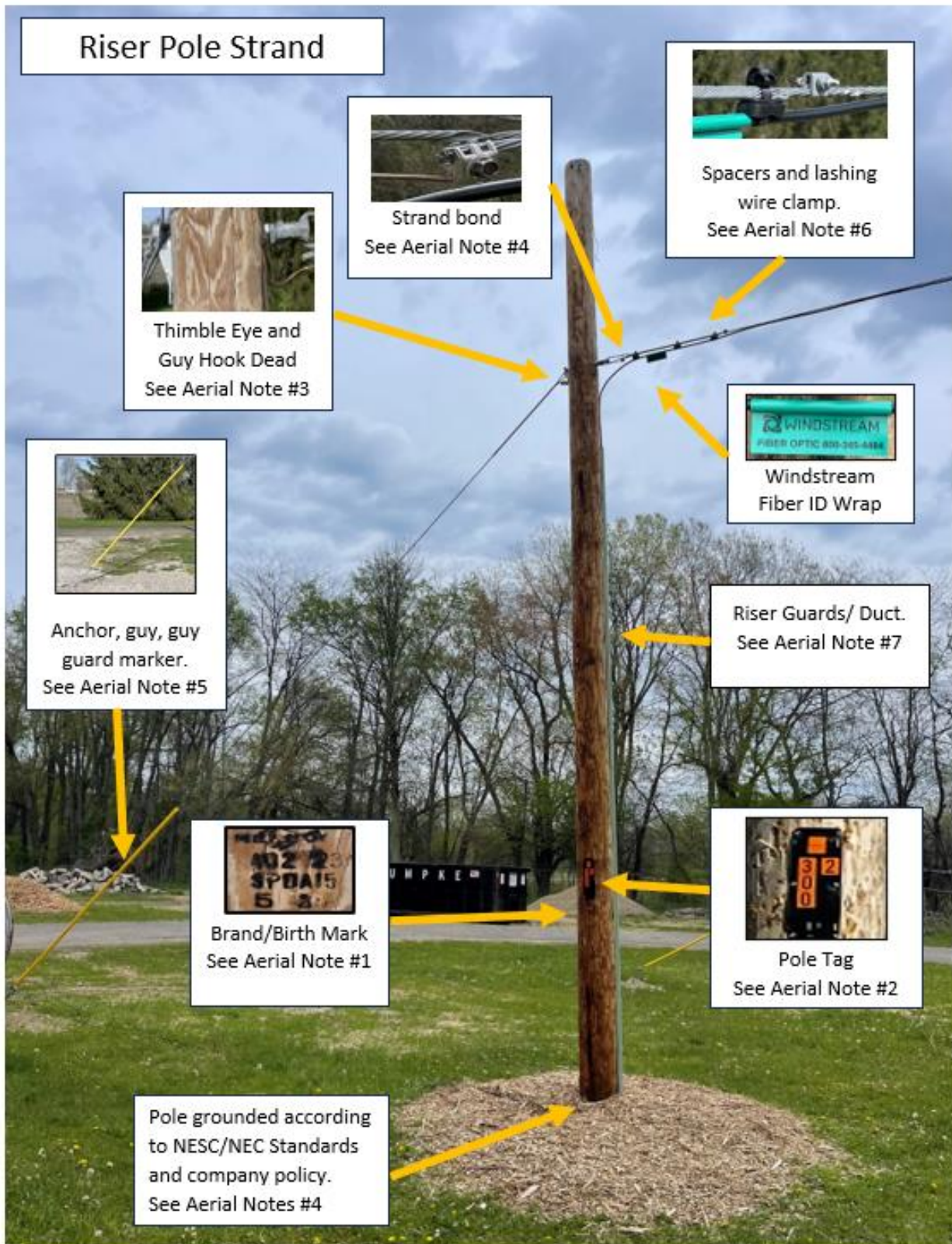


2. P-Clamps:

- a. Shims under wedges with fiber cable between it and sleeves.



Aerial - Stranded:



In-line Strand



Strand bonded and grounded.
See Aerial Notes #4



Spacers and lashing wire clamp.
See Aerial Notes #6



In line hardware
See Aerial Notes #8.



Windstream
Fiber ID Wrap



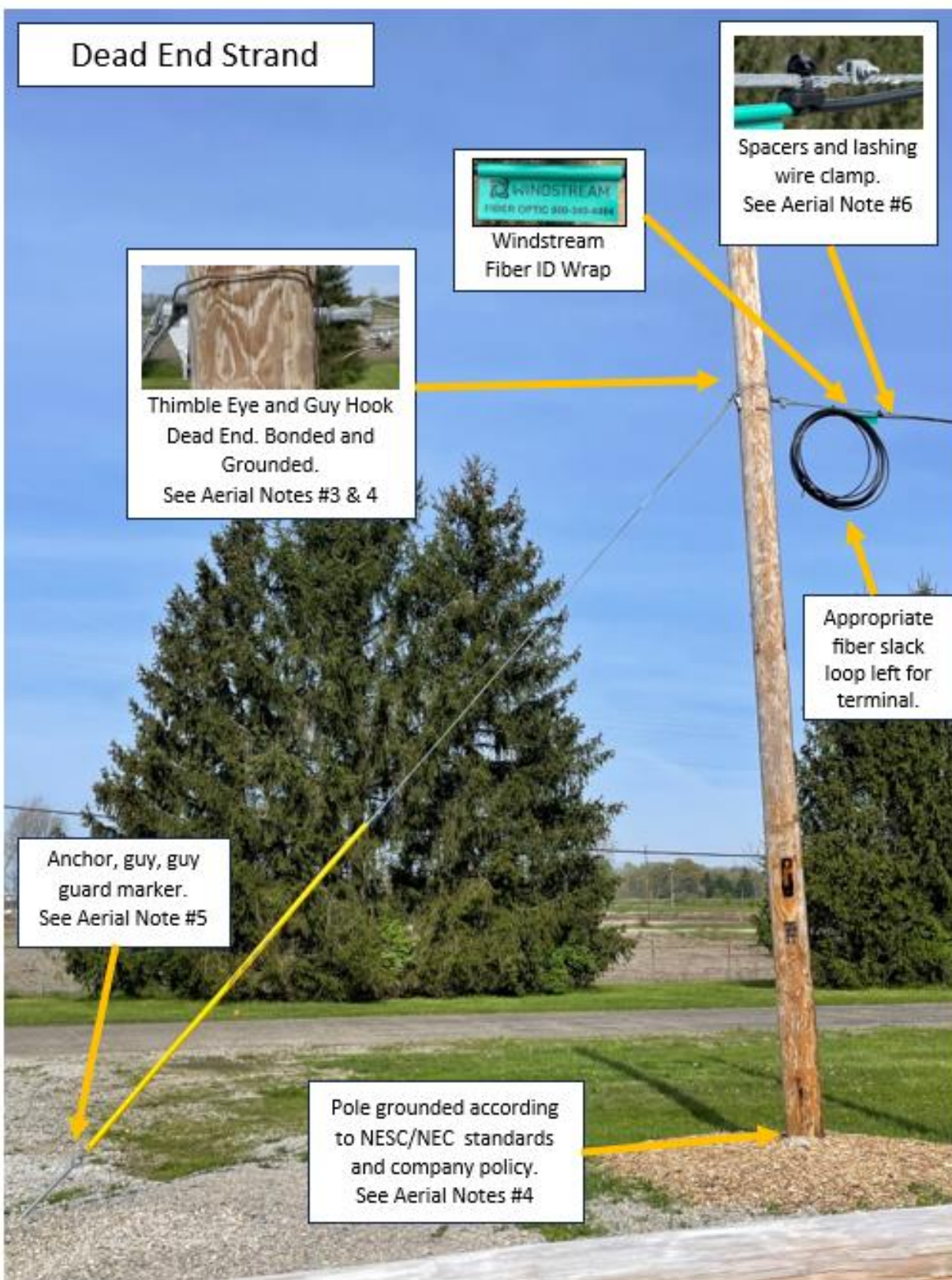
Brand/Birth Mark
See Aerial Note #1



Pole Tag
See Aerial Note #2

Pole grounded according to NESC/NEC standards and company policy.
See Aerial Notes #4





Dead End Strand



Thimble Eye and Guy Hook
Dead End. Bonded and
Grounded.
See Aerial Notes #3 & 4



Windstream
Fiber ID Wrap



Spacers and lashing
wire clamp.
See Aerial Note #6

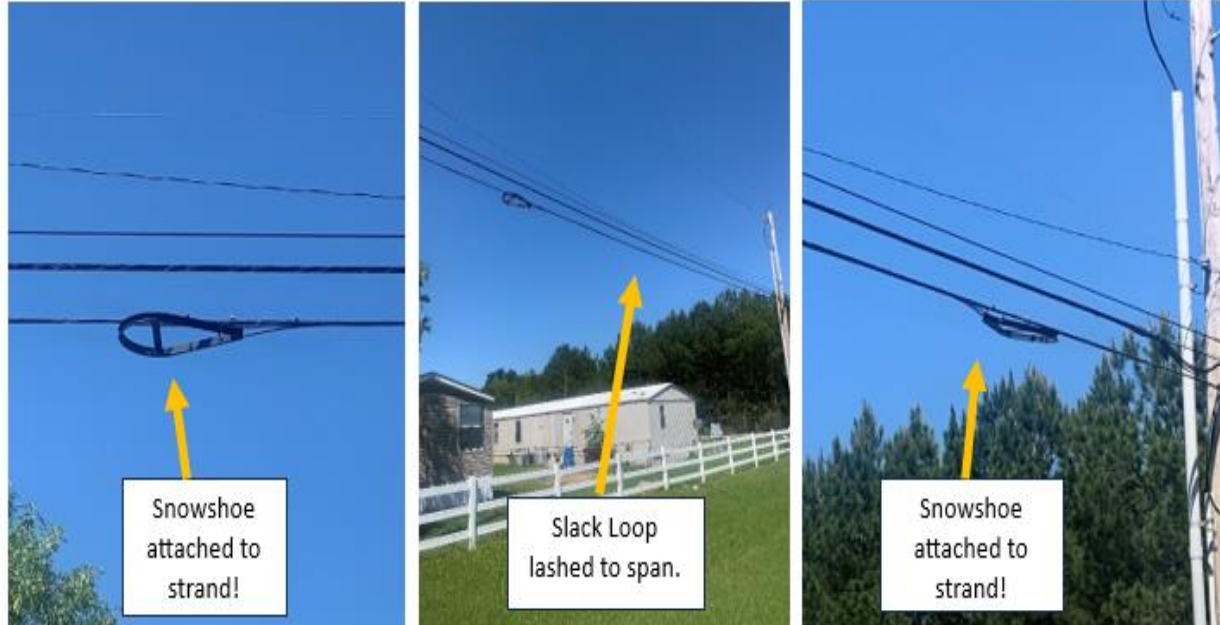


Appropriate
fiber slack
loop left for
terminal.

Anchor, guy, guy
guard marker.
See Aerial Note #5

Pole grounded according
to NESC/NEC standards
and company policy.
See Aerial Notes #4

Slack Loop and Snowshoes



Installation:

- Slack loops are to have "Snowshoes" installed on each end and connected to the strand with provided hardware.
- Fiber cable to be secured with Deltec strapping.
- Loop is to be lashed to strand.
- Best practice is to lash separately from home run fiber cable.
- Lashing wire clamps placed properly at each snowshoe.



Aerial Notes:

1. Brand / Birth Mark:

- Shows pole class, size, height and can be used to verify depth.
- Birth Mark Height is typically 10' ft from butt of pole.

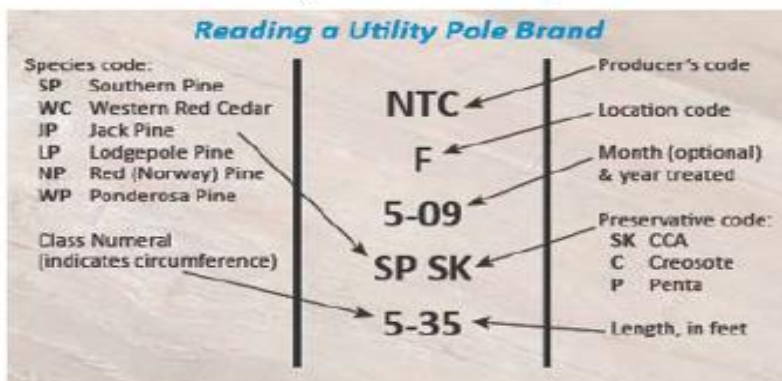
Metal Tag (Bellybutton) Method



Stamped (Burned or Branded) Method



Utility Pole Brand Legend



2. Pole Tag:

- a) Install vertical and shows lead, structure # and ownership.
- b) Good Time to check that pole is plumb



3. Hardware for Dead End:



- a. All attachment hardware secured in place at proper height.
- b. Guy hook directly against pole with no washer to prevent turning.
- c. Thimble Eye to have min. of 3 threads inside. Excess length cut off.
- d. Must have at least 1 nut between thimble eye and washer on span side.
- e. Washer squared up with pole.
- f. Eye turned up for inline and to the side if any side pull.
- g. Preform wraps installed correctly with tail protruding to connect ground.
- h. Strand not sagging. Tensioned correctly and matching existing utilities if available.
- i. Span to have proper clearance height over roads, sidewalks, etc...



4. Bonding / Grounding:

Strand bonded to MGN or pole ground wire using approved clamps and #6 AWG ground wire.

If MGN is not provided a vertical ground must be installed using a 5/8" in ground rod, connected to #6 AWG ground wire with approved clamp and ground wire ran to top of pole using ground staples.

Ground staples should only be 6"-8" inches apart, as high as a person can reach and then approx. 14" inches apart to top of pole.

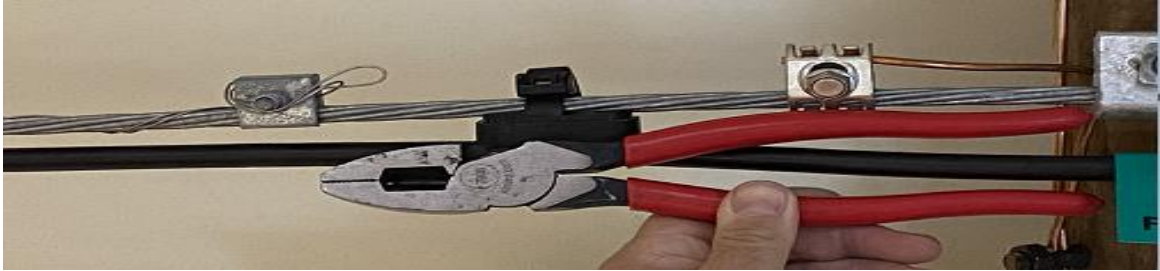


5. Anchor, Guy, and Guy Guard:

- a. Anchor placed in (ROW) Right of Way.
- b. Anchor not extending out from ground more than 12" inches.
- c. Anchor's lead/angle correct according to engineered specs.
- d. Guy guard markers installed over guy wire and secured with Deltec Strapping.
- e. Guy tensioned enough that pole is plumb or only slightly angled.
- f. Any overhead guys to have proper height clearance.



6. Spacers & Lashing Wire:



Inline:

- a. Lashing wire clamp should be installed 9" from suspension clamp and spacer should be placed 2" inside of lashing wire clamp.
- b. Lashing wire should be wrapped onto bare strand with a minimum of 2 wraps before attaching to lashing wire clamp.
- c. All Fiber optic cable should be double lashed and should be uniform and be neat in appearance.



Dead End:

- a. Spacers installed using Deltec strapping and locking heads every 12"-14" inches.
- b. Lashing wire clamp tight and placed at edge of preform wrap onto the strand with minimum of (2) wraps of wire around strand.

7. Riser guards/duct:

- a. Should fit fiber cable or innerduct.
- b. Cable straps placed from guards to strand at approx. 18" interval or less.



8. In- Line Hardware:

- a. Storm support strap: Used on 10m strand only.
- b. Straight through suspension clamp: For inline.
- c. Corner suspension clamp: For angled spans.
- d. Back side in-line hardware: Should be equipped with washer squared up.
- e. Back side corner pole hardware: Should have guy hook only with no washer.
- f. Correct length for through bolt: No more than 1.5 inches of bolt protruding out. past nut or min. of (3) threads showing. Bolt can be cut for proper length.



Buried Construction

Cable Depths:

NOTE: Depths outlined below are the standard placement depths for utility cables.

In ALL circumstances, placement crews should adhere to, and abide by, the standards set forth by the engineer, permitting entity, or suggested depth set forth by Windstream or the right of way owner/permittee.

Chart of Direct Buried Telecommunications Cabling Depths (RUS)

Copper Cabling	Minimum Depth (in)
Normal Soil Conditions	24
Ditch Crossing	36
*Solid Rock Placement	6
Customer Premise Placement (drop)	12

Fiber Optic Cabling	Minimum Depth (in)
Normal Soil Conditions	42
Ditch Crossing	42
*Solid Rock Placement	6
Customer Premise Placement (drop)	12

*Note: Solid rock placements require sand or select soil backfill 2" below and 4" above cabling prior to backfilling trenchline.



Route Markers:



Buried cable Route Markers (aka: "matchsticks") should be vertically level.

Matchsticks should have anchor bar installed to prevent post from being pulled out of ground and should be buried approx. 2' deep.



Restoration:

Bore Restoration:

Proper use of Tarp for Bore Pit Materials

Proper separation of SOD for reuse after job Completion.

Maximum effort should be made to save sod for backfilling & restoration.

- Pits are backfilled, tamped, & raked.
- Contractor Blend grass seed & straw placed to repair lawns where sod could not be saved.

Plow Restoration:



- Ensure caution tape is installed while plowing!!



- Simply drive over with tire to close trench line.
- Rake as needed to dress the area!!

Trench Restoration:



Trench kept clean of trash/debris & depth maintained.

Ensure caution tape is installed during backfilling operations!!

Trench is backfilled & raked.
(See Area Based Options Below)



Option 1 (Non-Residential Areas):

- Trench is left with 4-6" mound above grade for future ground settling.
- Trench line is raked/dressed neatly.

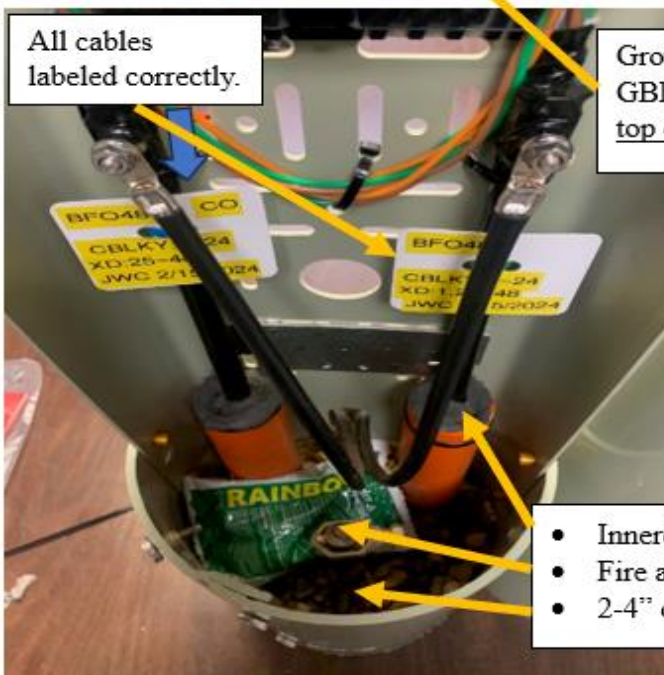


Option 2 (Residential Areas):

- Trench is backfilled, raked, & tamped approx. even with grade.
- Seed & straw should be placed over trench line.

Pedestals, Flowerpots & Fiber Distribution Hubs, Handholes

Pedestals:



Ground Rod connection visible & connected to GBB. Ground rod should be no higher than top of pedestal base.

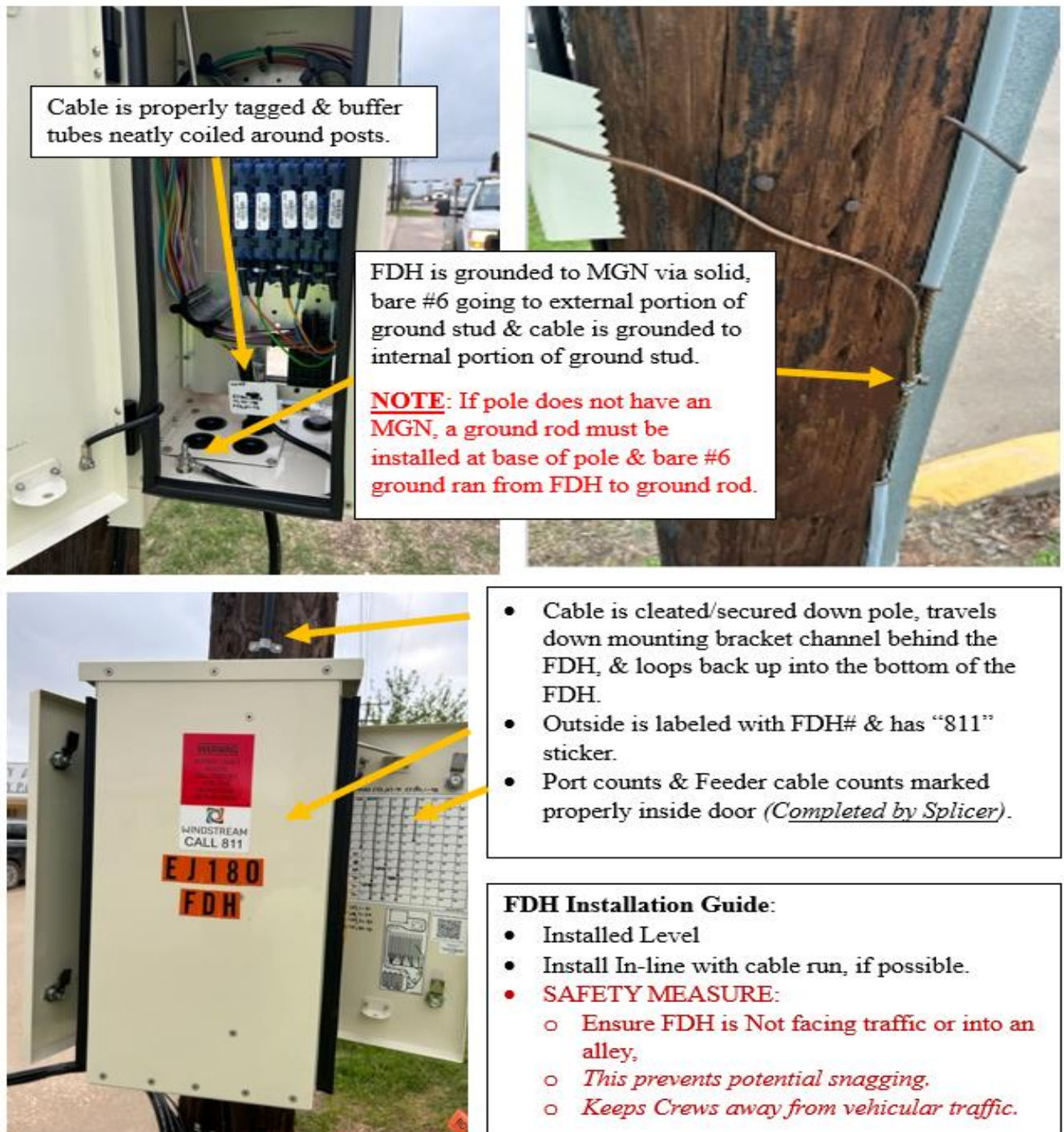
- Innerducts sealed & secured.
- Fire ant killer.
- 2-4" of pea gravel & moisture barrier installed.

Flowerpots:



Lid should be securely in place & level.

Pole Mounted – (FDH) Fiber Distribution Hub:



Pedestal Style – (FDH) Fiber Distribution Hub:



Cable secured via ty-wraps to pedestal .
Framing, is neat in appearance, and properly.
Labeled / tagged.

NOTE: Color-coded vinyl tape should be
used for marking cables instead of ty-wraps.
The color-coded ty-wraps are not UV
resistant and will rot/fall off.

Ground rod installed, #6 ground from ground
rod to GBB, & all cables grounded
individually to GBB.

NOTE: This pedestal is grounded using the
GBB supplied with the pedestal. Optionally,
the installer can ground everything to the
FDH supplied GBB, but a ground lug is
needed for the ground rod ground wire
connection (*Vertiv P/N P36597 or similar*).

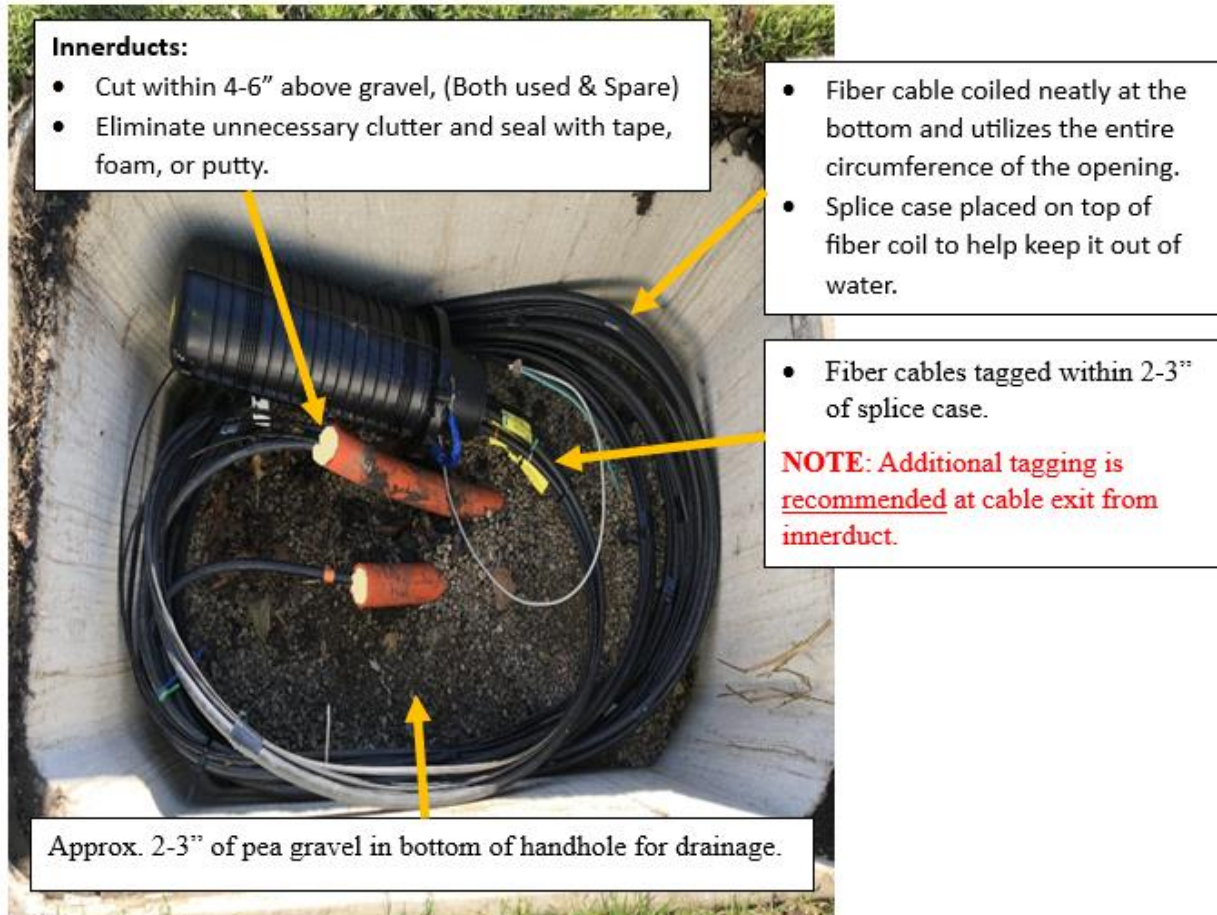


Proper amount of pea gravel (2-4") & Fire
Ant Killer is installed.

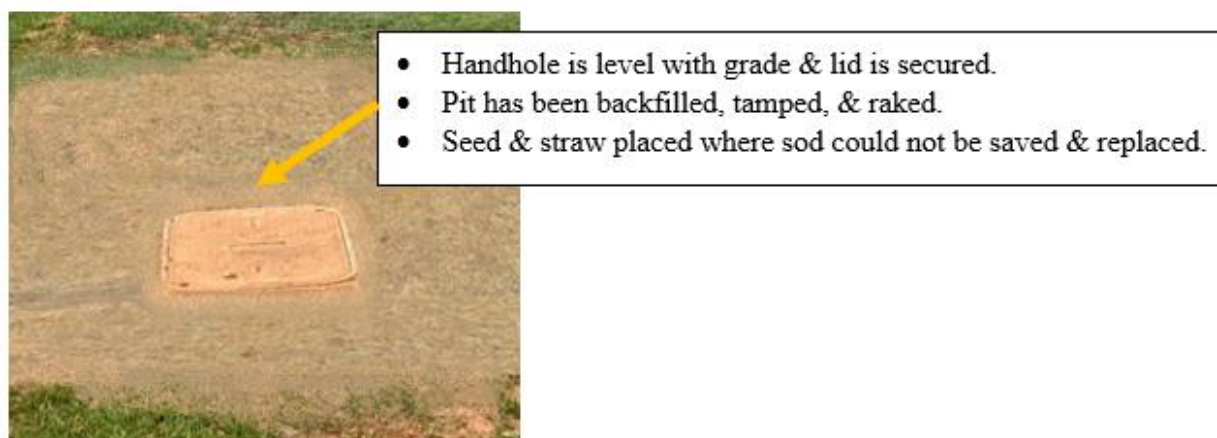
Pedestal is properly marked with FDH #
and "811" sticker.



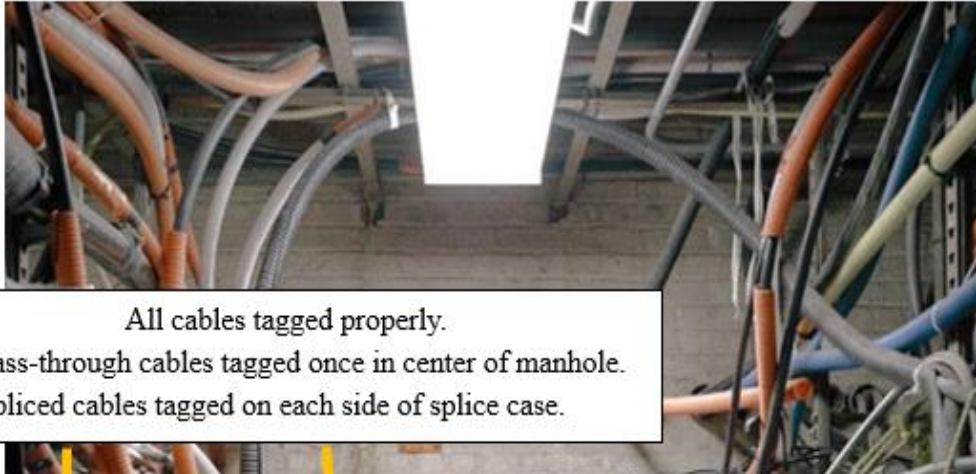
Handholes:



Handhole Restoration:



Manholes & Central Office (CO) Vaults:



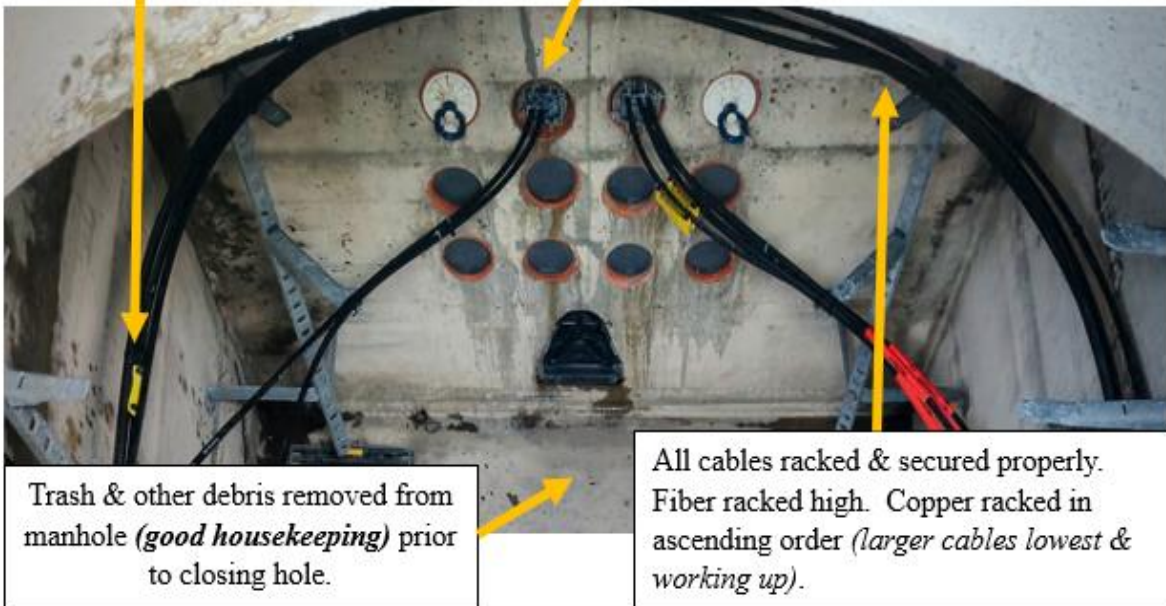
All cables tagged properly.

- Pass-through cables tagged once in center of manhole.
- Spliced cables tagged on each side of splice case.



All conduits/innerducts entering CO (or other buildings) are sealed in the feeding manhole/handhole and inside the CO/building.

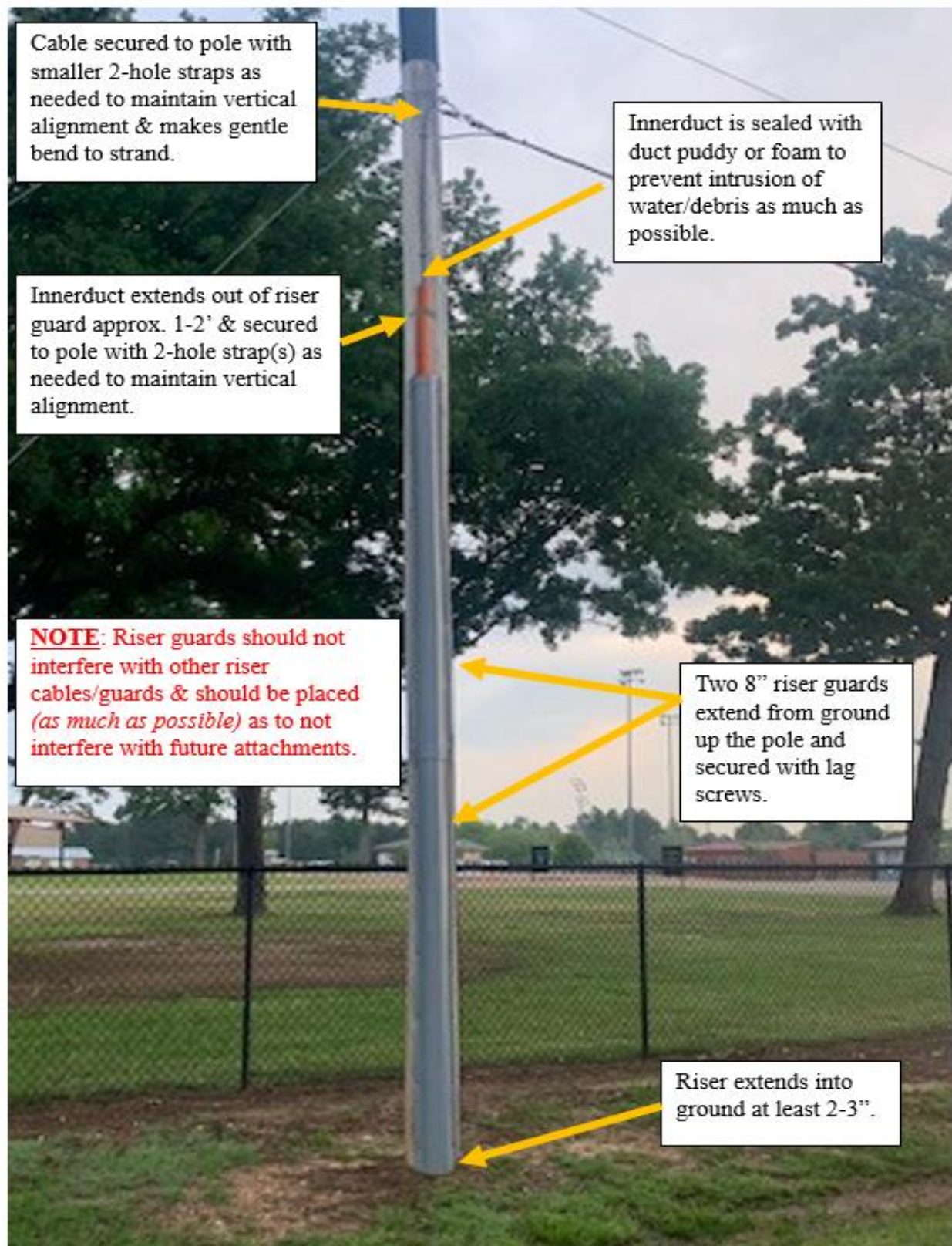
This includes used and spare conduits/innerducts.



Trash & other debris removed from manhole (*good housekeeping*) prior to closing hole.

All cables racked & secured properly. Fiber racked high. Copper racked in ascending order (*larger cables lowest & working up*).

Risers:

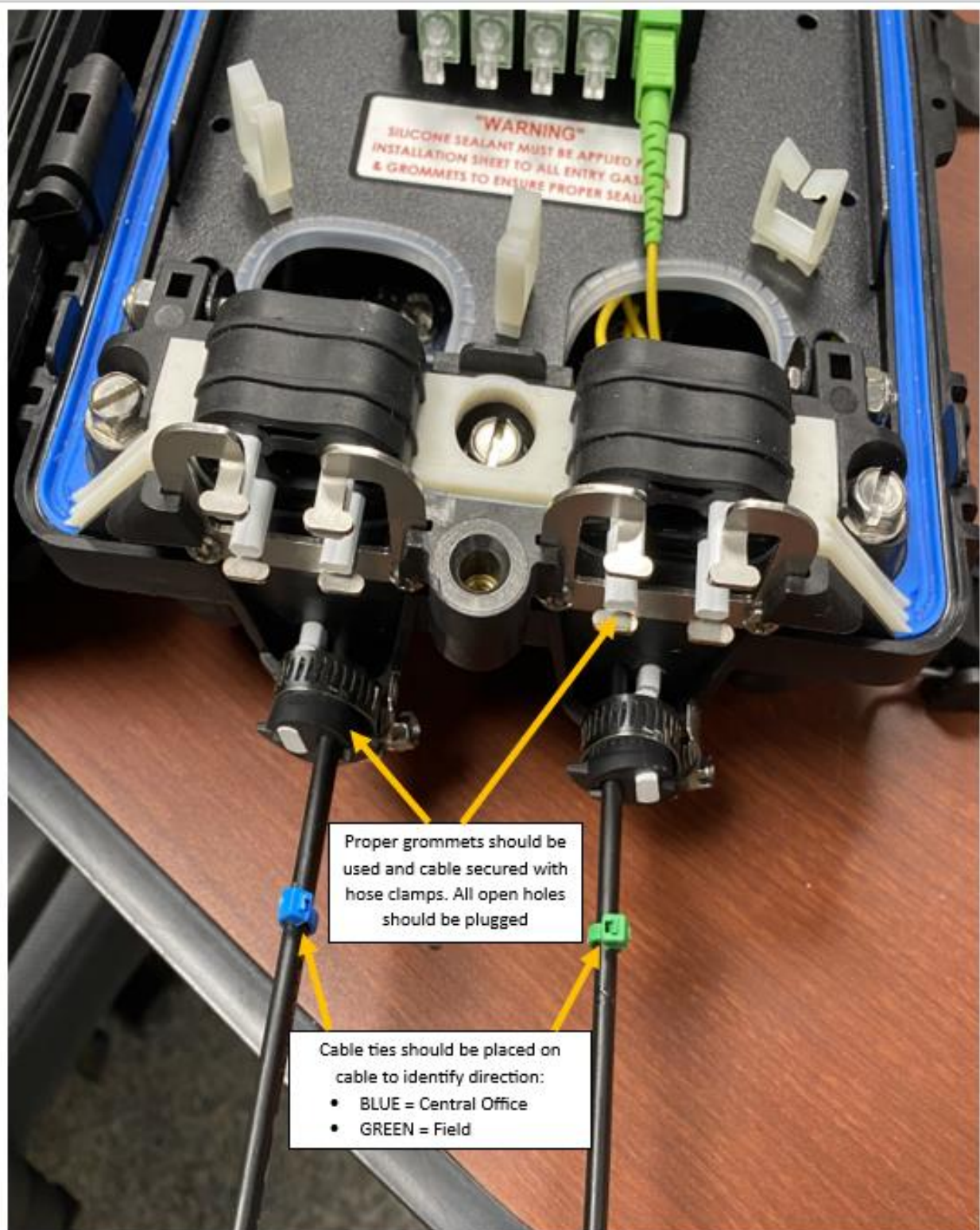


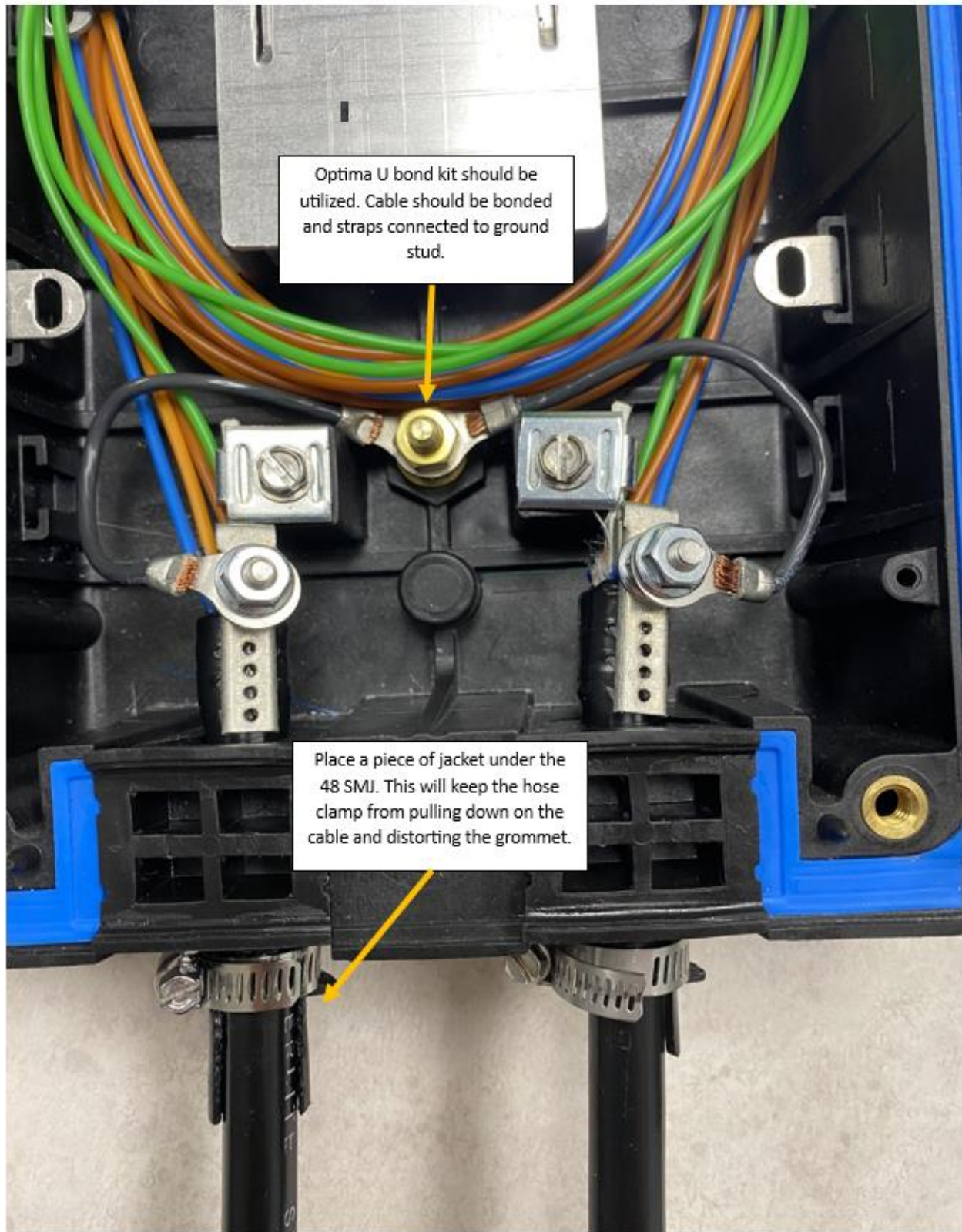
Splicing:

OPTIMA Splice Case:



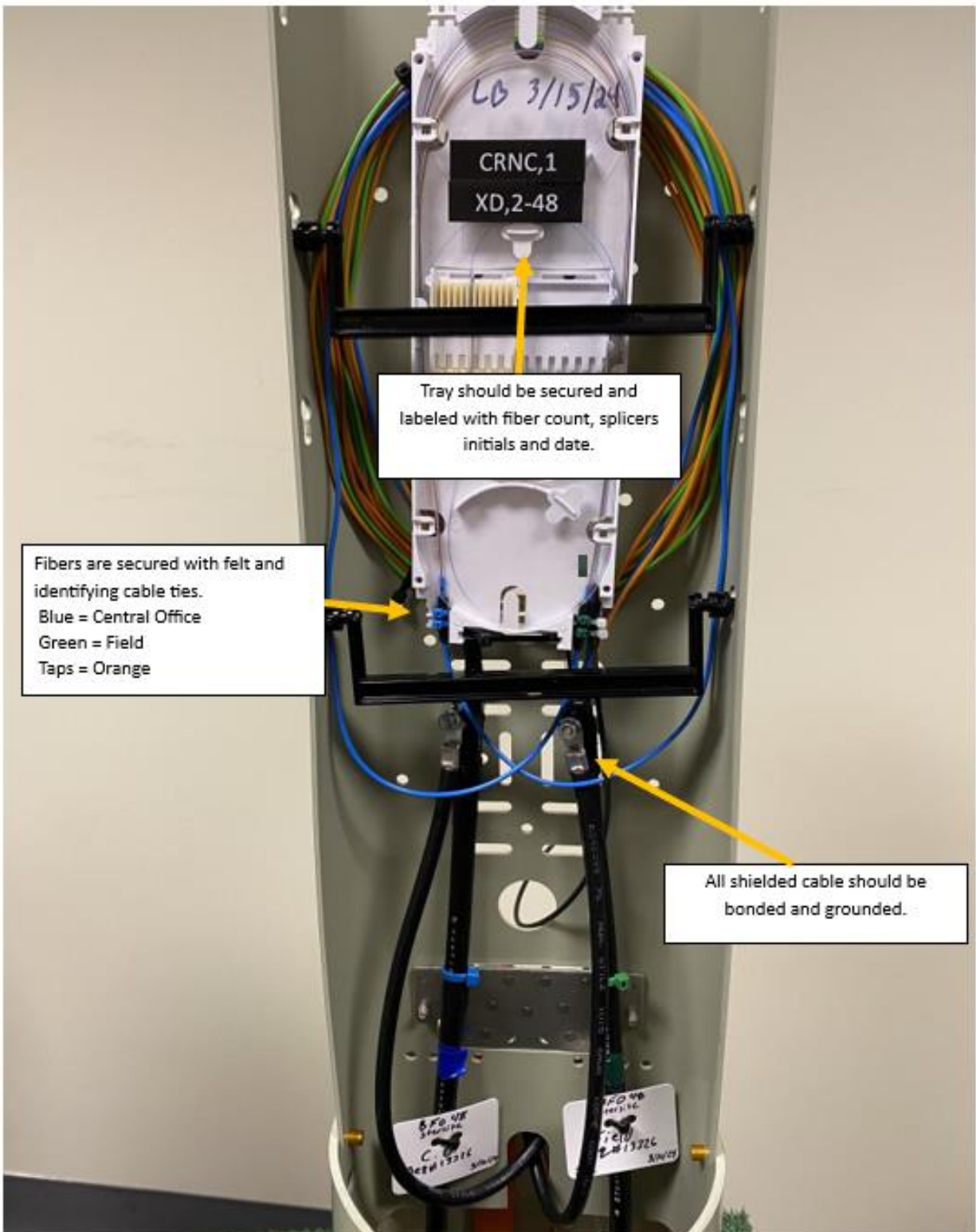


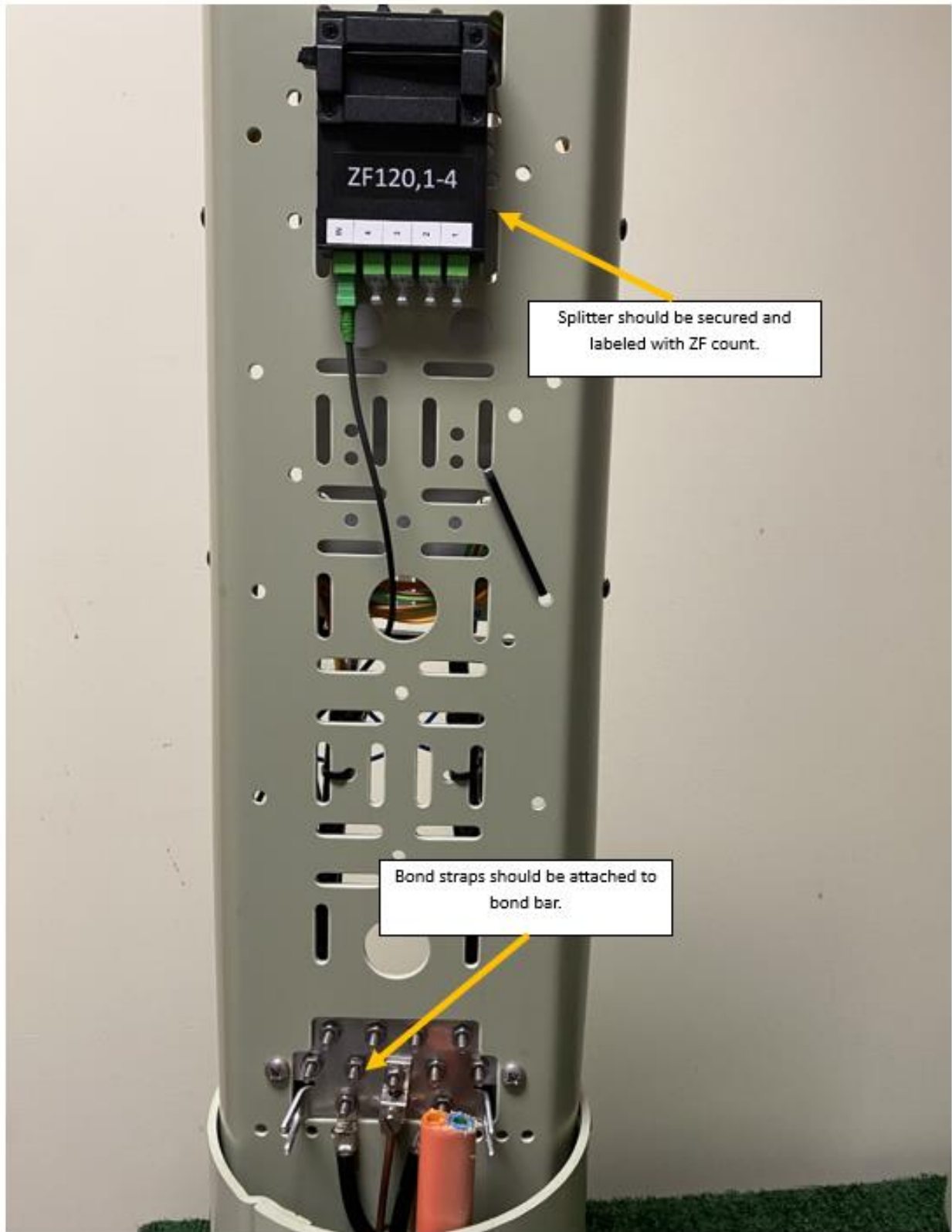


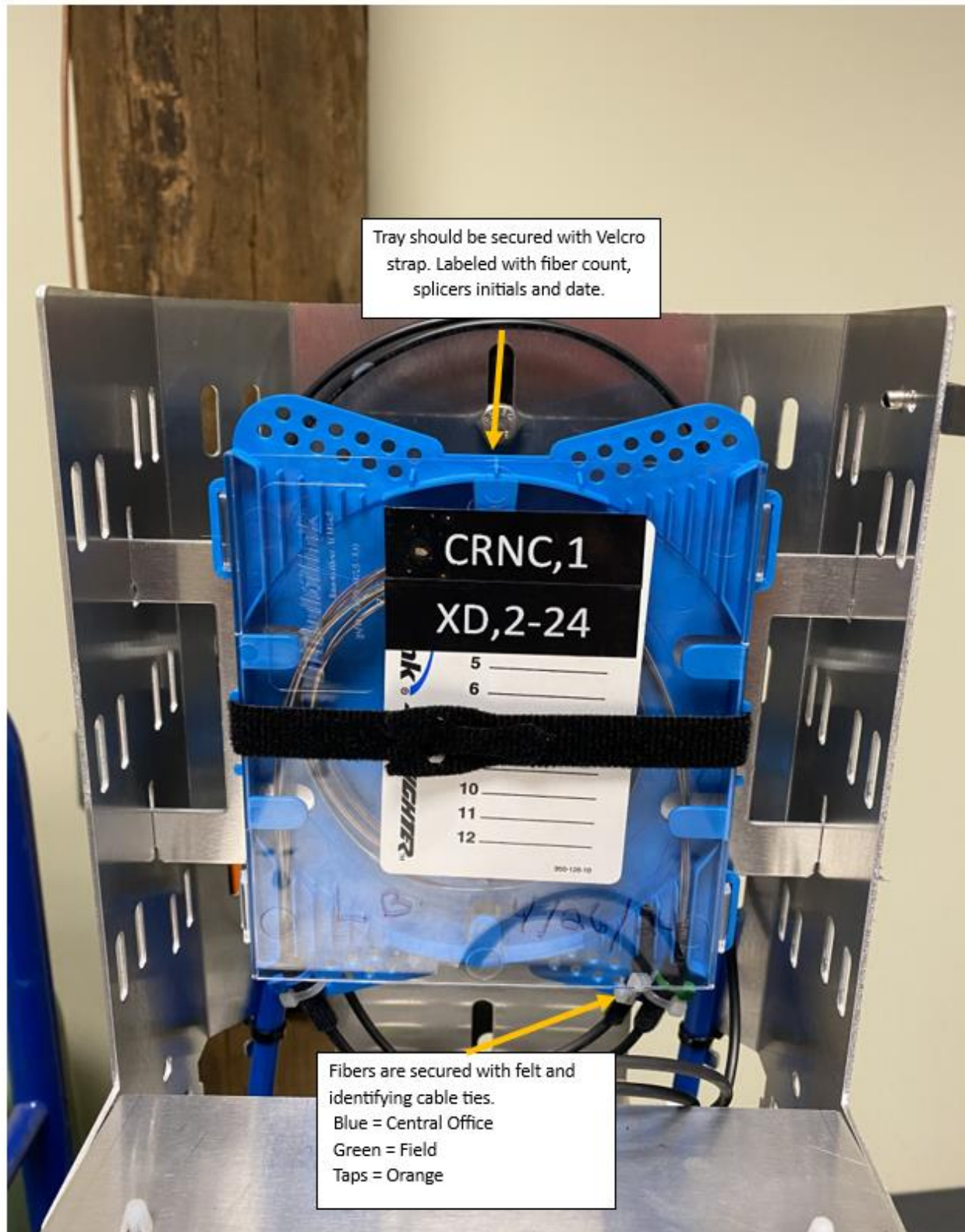




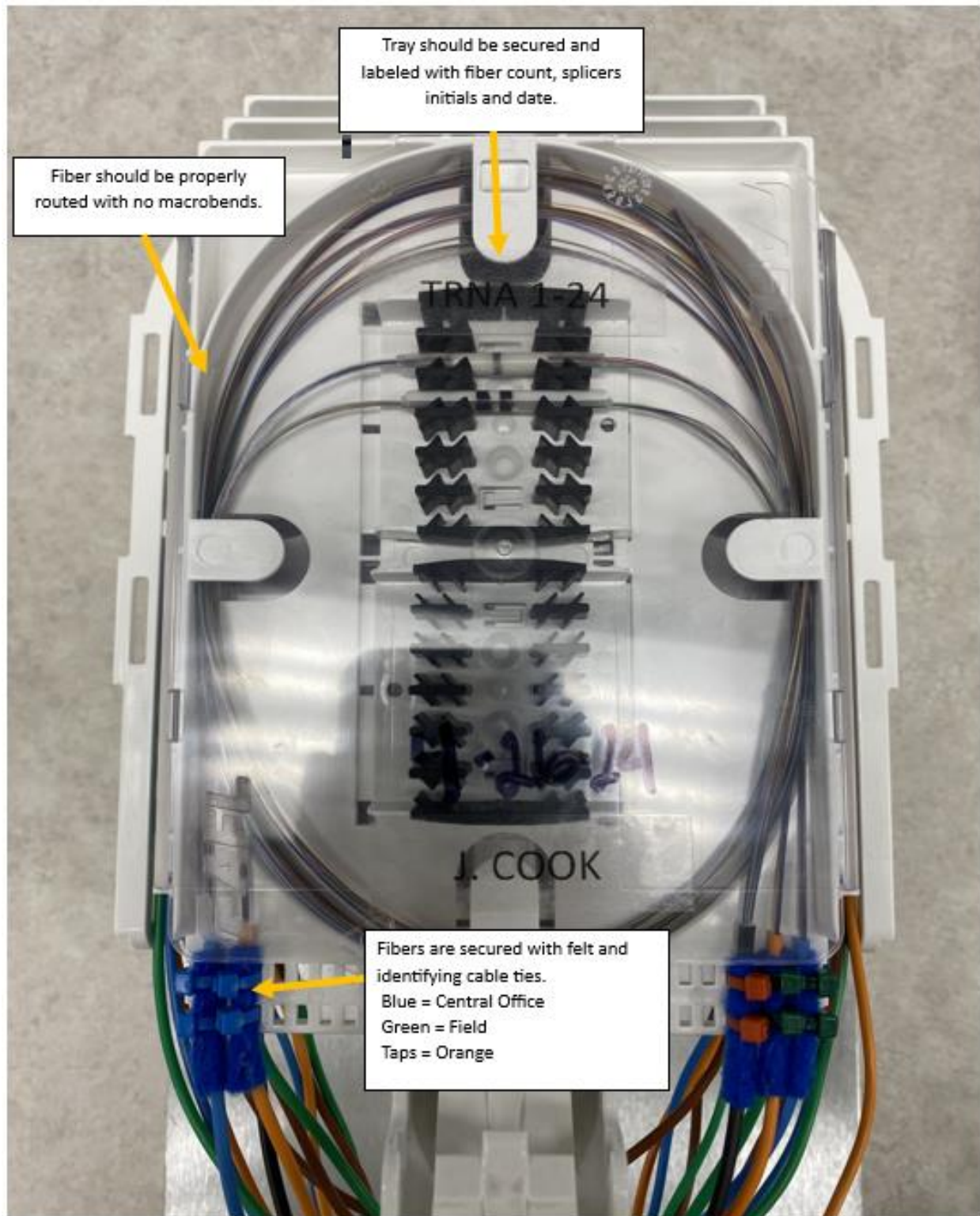
Splicing in Pedestal:

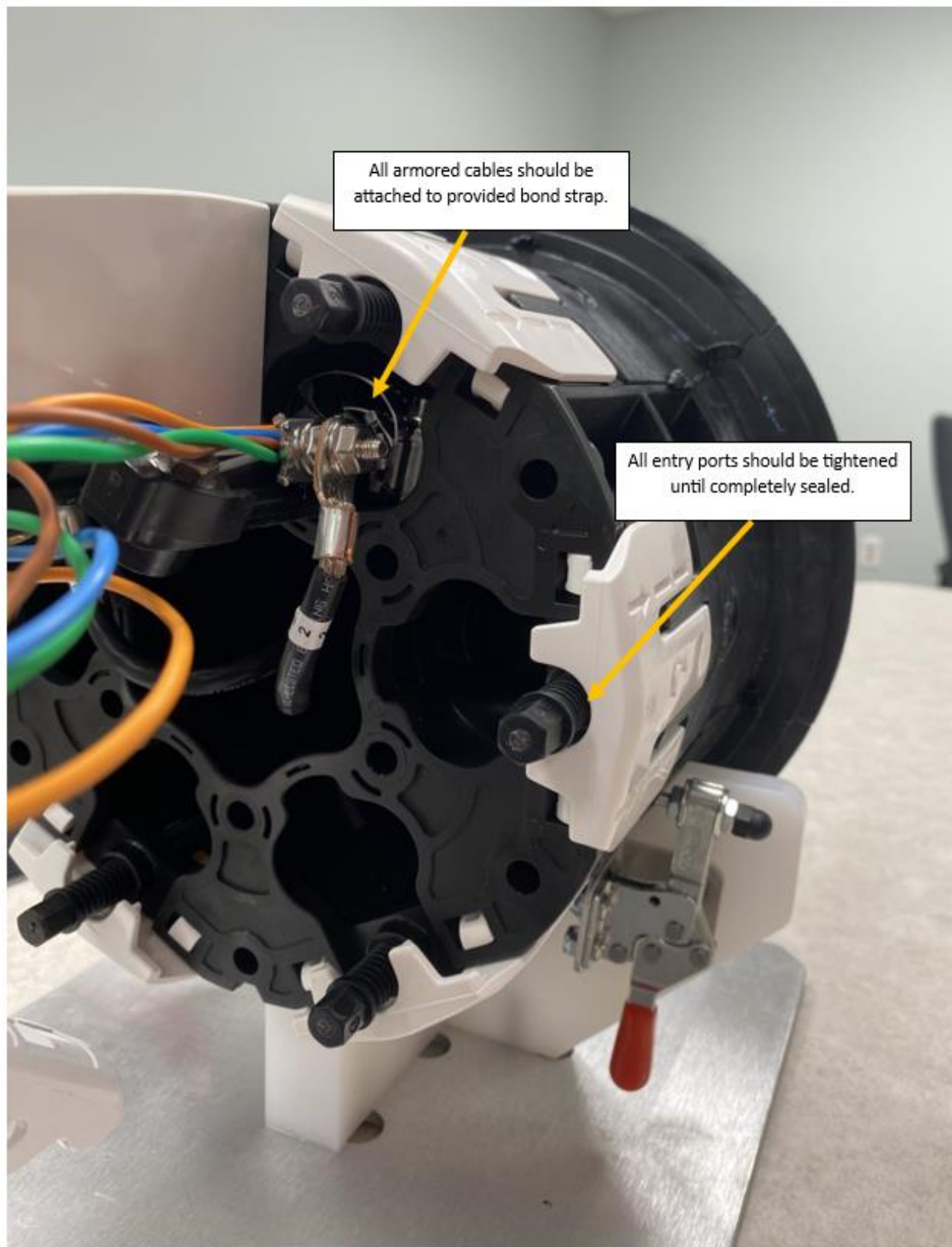


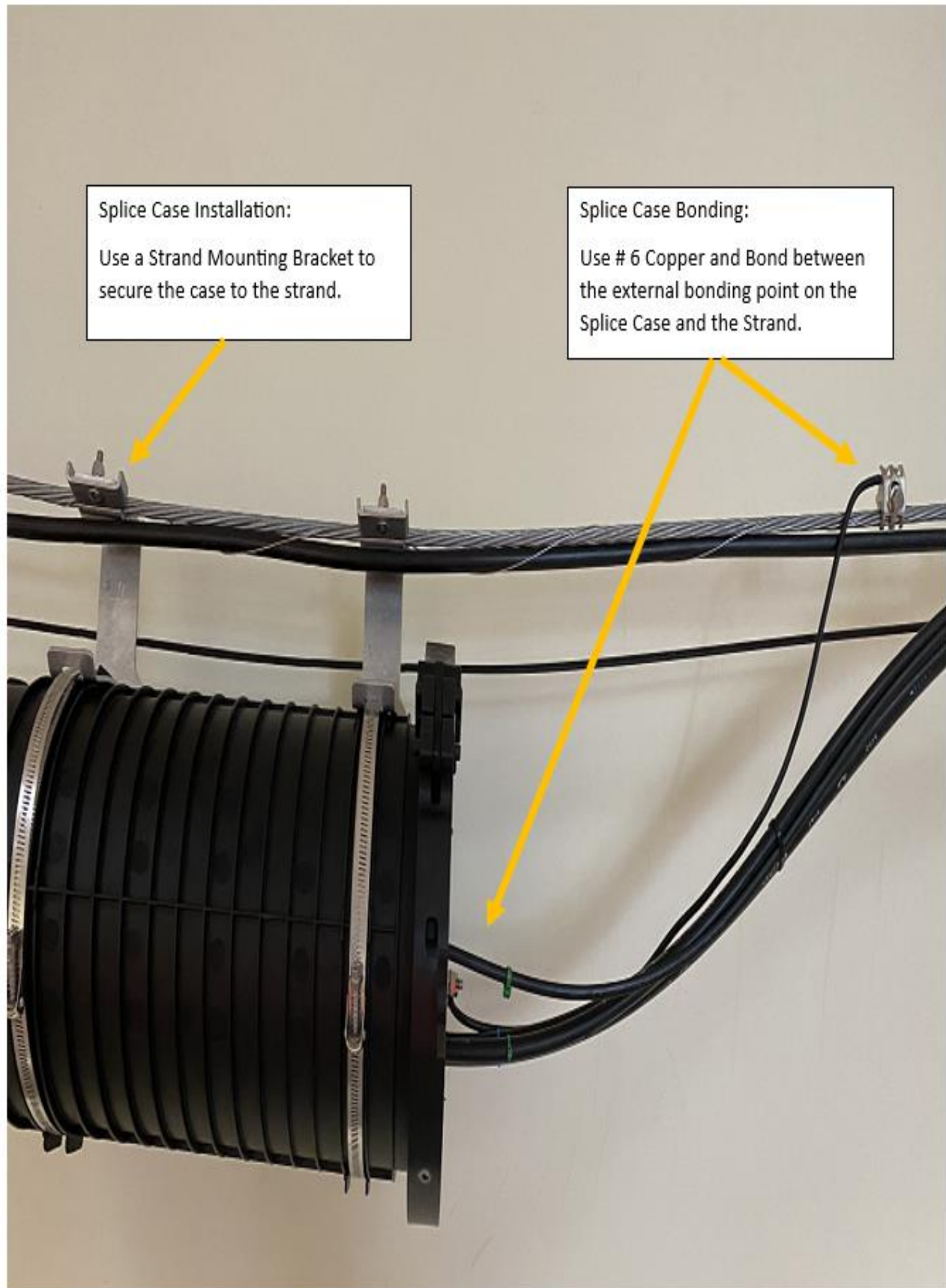




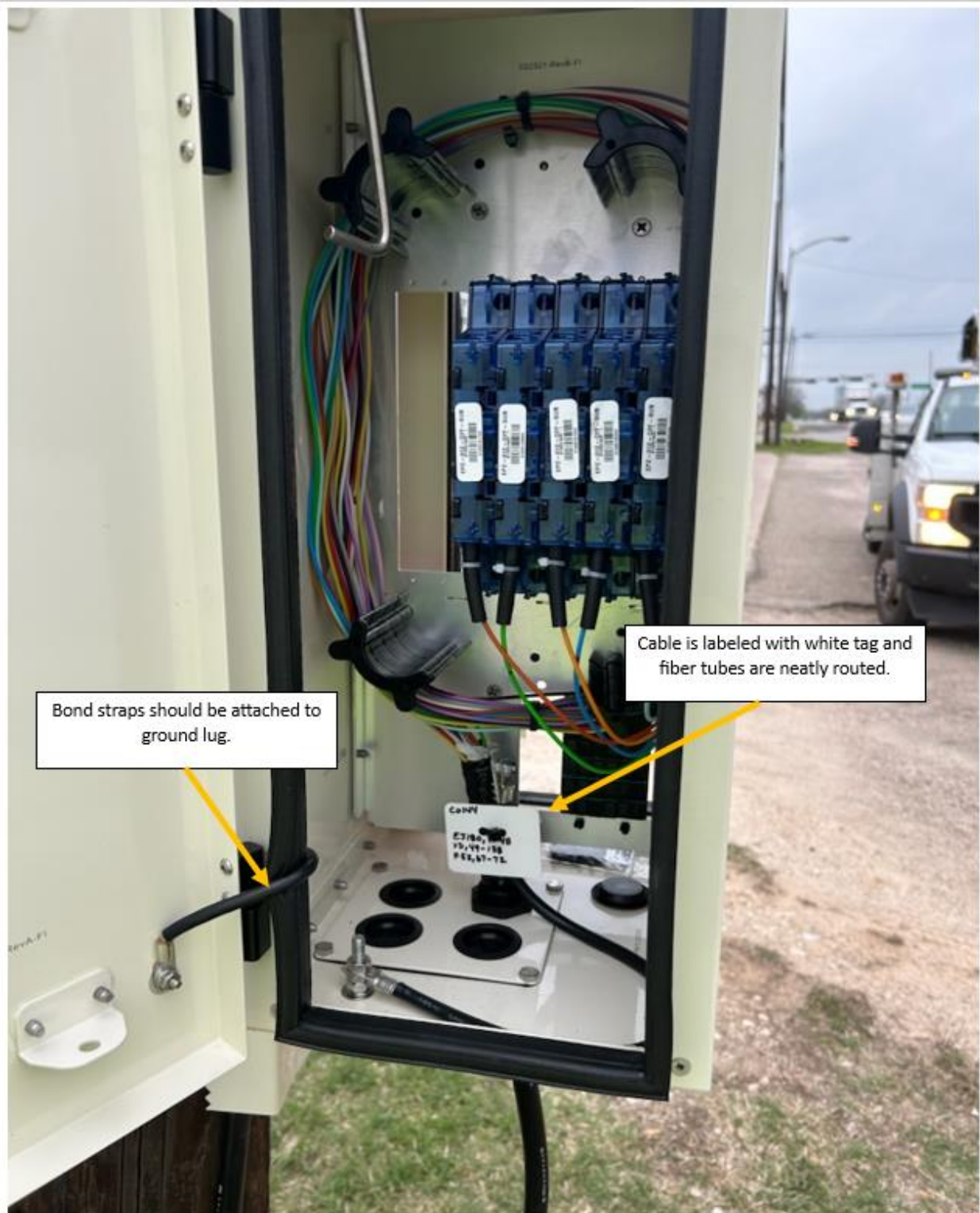
Splice Cases:

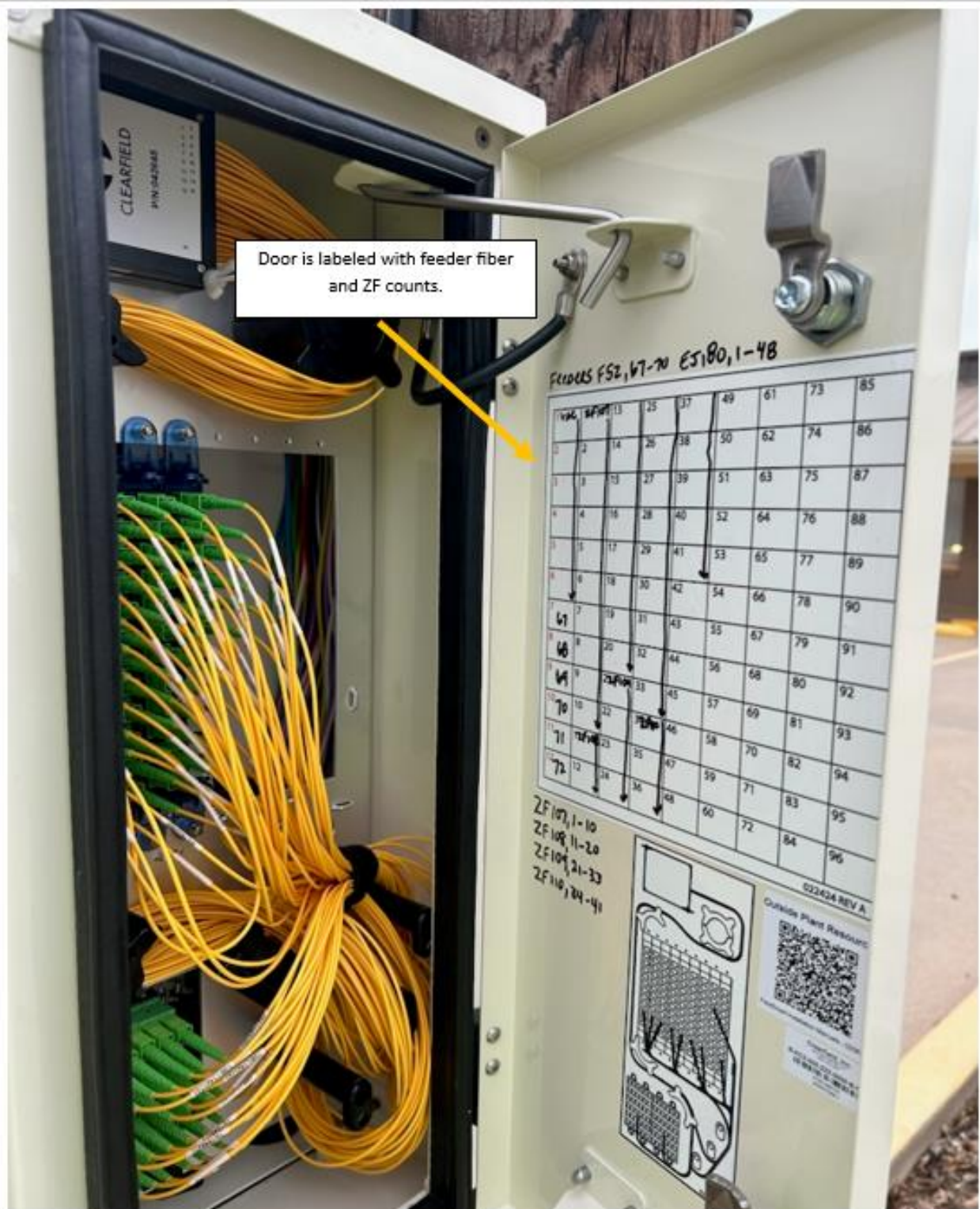






Fiber Distribution Hub (FDH)





Appendix-A: Inspection Forms _ Downloadable

AERIAL Inspection Form:



OSP-Aerial
Inspectionv1_fillable



OSP-Aerial
Inspection.xlsx

BURIED Inspection Form:



OSP-Buried
Inspection v1_fillabl



OSP- Buried
Inspection.xlsx